

houseCARL on Vortex — Shim Setup

houseCARL hard-requires Mod Organizer 2 because it reads load order from MO2's profile files. The workaround: build a small directory that looks like an MO2 instance, but with its files generated from (or symlinked to) Vortex's equivalents. houseCARL never talks to MO2 the program — only to four text files and the mod folders they reference — so a static shim is enough.

How the two managers map onto each other

Four things houseCARL needs, and where Vortex keeps the equivalent:

MO2 file	What it is	Vortex equivalent
ModOrganizer.ini	Instance config — names the game and points at the profile + mods folders.	None. You write this yourself once.
plugins.txt	Enabled .esp/.esm/.esl list (with * prefix for active).	%LOCALAPPDATA%\Skyrim Special Edition\plugins.txt
loadorder.txt	Plugin names in load order, no prefix.	%LOCALAPPDATA%\Skyrim Special Edition\loadorder.txt
modlist.txt	Mod folder names with +/- prefix, bottom-to-top = highest priority.	Generated from a listing of Vortex's staging folder.
mods/<ModName>/	Each mod's actual files (plugins, meshes, textures, scripts).	Vortex's staging folder. Each subfolder is one installed mod.

Vortex deploys to the Skyrim Data folder via hard links, so the plugin files in the staging folder are the same bytes as the deployed files. Reading from staging is correct.

Prerequisites — find these paths first

1. Find your Vortex staging folder. In Vortex: Settings → Mods, look for "Base Path" or "Mod Staging Folder." It's typically something like D:\Games\Vortex Mods\skyrimse\ or C:\Users\<you>\AppData\Roaming\Vortex\skyrimse\mods\. Whatever it is, write it down.
2. Confirm your Skyrim AppData files exist. Open Run (**Win+R**) and paste: %LOCALAPPDATA%\Skyrim Special Edition. You should see plugins.txt and (likely) loadorder.txt.
3. Confirm your Skyrim install folder. The Steam install of Skyrim SE — needed so houseCARL knows where the game itself is.
4. Install .NET 9 ASP.NET Core Runtime if you haven't already, and install houseCARL itself via houseCARL-Setup.exe with Claude Code or Codex as the host.

Build the shim

Pick a location for the shim. Anywhere writable works — example uses C:\houseCARL-Shim\.

Step 1 — Create the directory tree

Open PowerShell as your normal user (admin only needed for symlinks — see note below). Replace the paths in the first three lines with yours, then run:

```
# --- EDIT THESE THREE LINES ---
$Shim = "C:\houseCARL-Shim"
$VortexStaging = "D:\Vortex Mods\skyrimse"          # your Vortex staging folder
$SkyrimPath = "C:\Program Files (x86)\Steam\steamapps\common\Skyrim Special Edition"
# -----

$AppData = "$env:LOCALAPPDATA\Skyrim Special Edition"
New-Item -ItemType Directory -Force -Path "$Shim\profiles\Default" | Out-Null
New-Item -ItemType Directory -Force -Path "$Shim\overwrite" | Out-Null
```

Step 2 — Symlink the mods folder to Vortex staging

Run PowerShell **as Administrator** for this one command (mklink needs it):

```
cmd /c mklink /D "$Shim\mods" "$VortexStaging"
```

If you can't or don't want to elevate, you can use a junction instead, which doesn't need admin:

```
cmd /c mklink /J "$Shim\mods" "$VortexStaging"
```

Either way, you should now see C:\houseCARL-Shim\mods\ listing every folder from your Vortex staging.

Step 3 — Wire up plugins.txt and loadorder.txt

Copy them in (refresh by re-running this whenever your load order changes), or symlink for always-fresh:

Option A — copy (simplest, run again after load-order changes):

```
Copy-Item "$AppData\plugins.txt" "$Shim\profiles\Default\plugins.txt" -Force
Copy-Item "$AppData\loadorder.txt" "$Shim\profiles\Default\loadorder.txt" -Force
-ErrorAction SilentlyContinue
```

Option B — symlink (admin) so they stay in sync:

```
cmd /c mklink "$Shim\profiles\Default\plugins.txt" "$AppData\plugins.txt"
cmd /c mklink "$Shim\profiles\Default\loadorder.txt" "$AppData\loadorder.txt"
```

Note: if loadorder.txt doesn't exist in AppData (some Skyrim setups only write plugins.txt), generate it from plugins.txt:

```
Get-Content "$AppData\plugins.txt" |
  Where-Object { $_ -and -not $_.StartsWith("#") } |
  ForEach-Object { $_.TrimStart("*") } |
  Set-Content "$Shim\profiles\Default\loadorder.txt"
```

Step 4 — Generate modlist.txt from the staging folder

MO2's modlist.txt has one mod per line, prefix + for enabled or - for disabled, and it's read bottom-to-top (last line wins for loose-file conflicts). Vortex doesn't have a direct equivalent priority order — its conflict winners come from deployment rules. As a workable approximation, list every staged mod as enabled, and order them roughly by mtime (most recently installed = highest priority, which is a rough proxy):

```
$mods = Get-ChildItem -Directory "$VortexStaging" | Sort-Object LastWriteTime
$lines = $mods | ForEach-Object { "+" + $_.Name }
Set-Content -Path "$Shim\profiles\Default\modlist.txt" -Value $lines
```

Better option if you care about loose-file winners: open Vortex → Mods, sort by your conflict rules, export the list, and order modlist.txt to match. For most use (record-level work, crash diagnosis), the rough order is fine — plugin-level conflicts are resolved by plugins.txt, which is exact.

Step 5 — Write ModOrganizer.ini

Save this as C:\houseCARL-Shim\ModOrganizer.ini, with your Skyrim path filled in:

```
[General]
gameName=Skyrim Special Edition
selected_profile=@ByteArray(Default)
gamePath=@ByteArray(C:/Program Files (x86)/Steam/steamapps/common/Skyrim Special Edition)
```

```
[Settings]
base_directory=C:/houseCARL-Shim
mod_directory=C:/houseCARL-Shim/mods
profiles_directory=C:/houseCARL-Shim/profiles
overwrite_directory=C:/houseCARL-Shim/overwrite
```

Note the forward slashes — MO2's ini convention, and Mutagen handles both, but match this format to be safe. The @ByteArray(. . .) wrapping is a Qt artifact; keep it on the two values shown.

Step 6 — Point houseCARL at the shim

Start Claude Code (or Codex), and when houseCARL asks for the MO2 instance folder on first use, give it:

```
C:\houseCARL-Shim
```

From the description, it persists across sessions. Ask it to list a few records or read a known plugin's contents to confirm it's reading correctly.

Handing patches back to Vortex

houseCARL writes each patch as a new mod folder under mods/ named houseCARL - <name>/. With the shim, that lands inside Vortex's staging folder. Vortex won't auto-pick-up arbitrary new folders, so you have two clean options:

5. **Manual mod from disk:** In Vortex, drag the new houseCARL - <name>/ folder onto the Mods page, or use Mods → Install From File (zip it first), then deploy. This is the safest path because Vortex tracks it normally.

6. **Out-of-band test:** To try a patch without committing it to Vortex, copy the .esp from the houseCARL output folder into your Skyrim Data folder and enable it in plugins.txt directly. Quick to test, easy to undo, but Vortex won't know about it — fine for short experiments, not for keeps.

Best practice: redirect houseCARL's output to a folder Vortex isn't watching, then promote good patches into Vortex. Easy way to do that: put a different folder behind mods/ — e.g., point the mods symlink at a directory that contains symlinks to each Vortex mod plus a writable houseCARL-Output/ subfolder. More setup, but it keeps Vortex's staging clean.

Caveats and gotchas

- **Loose-file conflict order is approximate.** MO2 modlist.txt order = loose-file priority. Vortex resolves loose files at deployment by rules, not by a single priority list. For record-level work (plugin conflicts), this doesn't matter — plugins.txt is authoritative for both. For things like which textures or scripts loaded, the shim will guess.
- **Refresh after load-order changes.** If you used Option A (copy) for plugins.txt/loadorder.txt, re-run the copy whenever you change your load order in Vortex. If you used Option B (symlink), no action needed.
- **Refresh after installing new mods.** New Vortex mods automatically appear under mods/ via the symlink, but they won't appear in modlist.txt until you regenerate it. Re-run the Step 4 PowerShell.
- **Plugin file location inside mod folders.** MO2 expects each mod folder to have a flat layout (plugin.esp at the folder root). Vortex staging mods do too — provided they were installed correctly. If a mod has its files nested one level too deep, houseCARL won't find the plugin. Same problem exists in Vortex itself (*no .esp found* warning), so anything that works in Vortex will work for houseCARL.
- **BSA archives.** houseCARL drives BSArch over the mod folders. Vortex-staged mods that ship BSAs will work fine here — same files.
- **Hard-link awareness.** Because Vortex hard-links from staging to Data, writes to files inside staging propagate to Data and vice versa. houseCARL only writes to new mod folders it creates, so it won't stomp on anything, but be aware if you ever edit files manually under the shim.
- **Crash logging.** houseCARL reads CrashLogger / Papyrus logs from their standard locations (Documents\My Games\Skyrim Special Edition\SKSE\). That's identical regardless of mod manager — full crash diagnosis lights up immediately.

Migration path if it goes sideways

If the shim approach starts producing wrong load-order winners on a complex conflict — most likely with mods that override loose files in conflicting ways — the cleanest fix is what the earlier crash work already pointed at: a parallel MO2 instance dedicated to authoring. The Nexus comment you saw described exactly that split: MO2 instance for mod authoring (where houseCARL runs first-class), Vortex for your playable install (which doesn't change). You're not migrating away from Vortex; you're adding a small MO2 setup next to it that points at the same game folder, with houseCARL pointed at MO2 for the authoring tasks where the shim's approximations would bite.

Either approach works. Start with the shim — it's an hour of setup against your existing install. Move to a parallel MO2 instance if you hit a case where the shim's loose-file ordering matters.